

The Brain in the Body

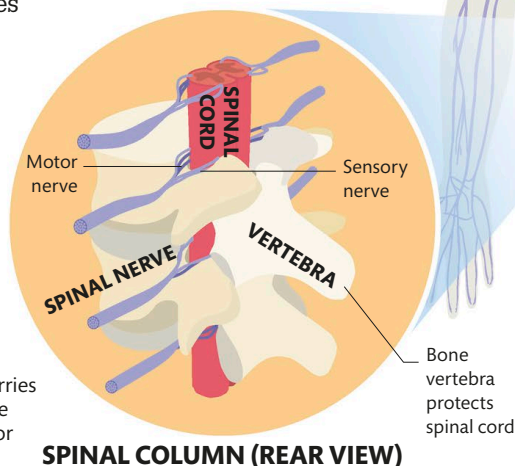
The brain is the primary component of the human body's nervous system, which coordinates the actions of the body with the sensory information it receives.

The nervous system

The two main parts of the nervous system are the central nervous system (CNS) and the peripheral nervous system. The CNS is made up of the brain and the spinal cord, a thick bundle of nerve fibers that runs from the brain in the head to the pelvis. Branching out from this is the peripheral system, a network of nerves that permeates the rest of the body. It is divided according to function: the somatic nervous system handles voluntary movements of the body, while the autonomic nervous system (see opposite) handles involuntary functions.

Spinal nerves

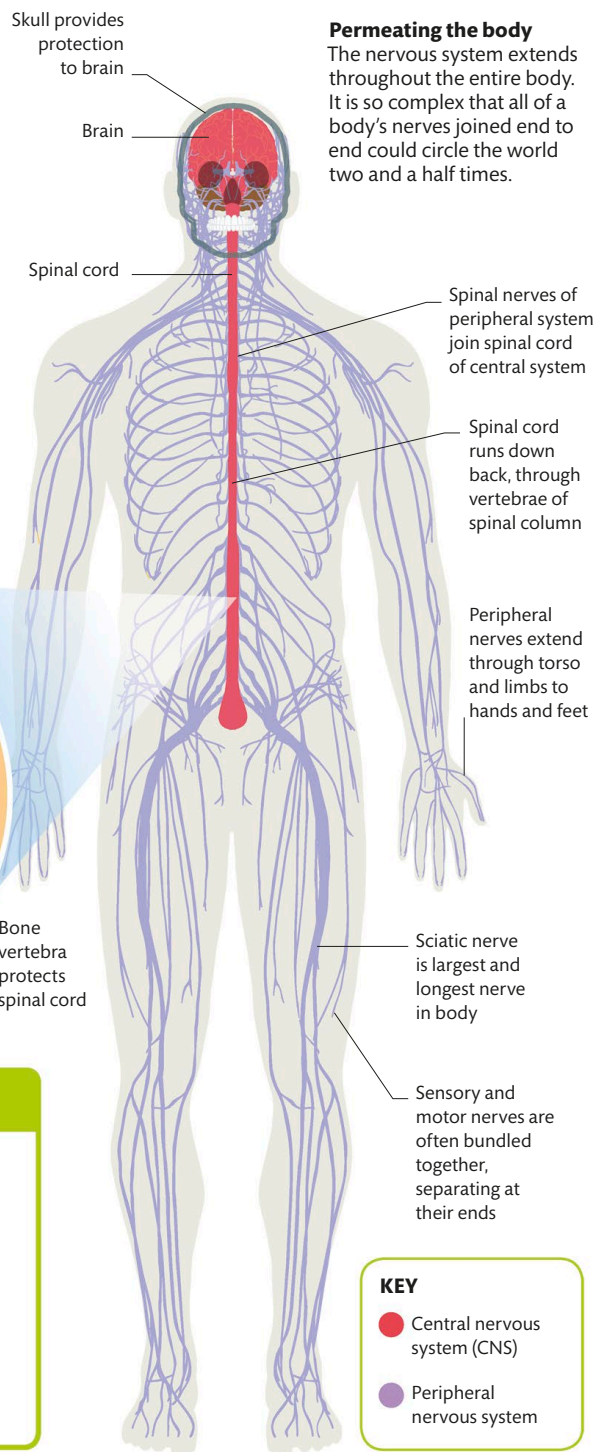
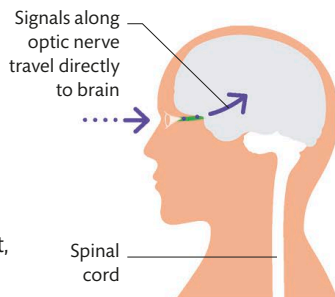
Most peripheral nerves connect to the CNS at the spinal cord and split as they connect. The rear branch carries sensory data to the brain; the forward branch carries motor signals back to the body.



SPINAL COLUMN (REAR VIEW)

CRANIAL NERVES

Within the peripheral system, 12 cranial nerves connect directly to the brain rather than the spinal cord. Most link to the eyes, ears, nose, and tongue and are also involved in facial movements, chewing, and swallowing, but the vagus nerve links directly to the heart, lungs, and digestive organs.



Permeating the body

The nervous system extends throughout the entire body. It is so complex that all of a body's nerves joined end to end could circle the world two and a half times.

KEY

- Central nervous system (CNS)
- Peripheral nervous system